Task 3: Crop Recommendation System Using Machine Learning

Introduction

Agriculture plays a vital role in the economy, and selecting the right crop based on environmental conditions is essential for maximizing yield. Machine Learning can help farmers make data-driven decisions by analyzing soil and climate parameters.

Objective

The objective of this task is:

- To build a crop recommendation system using machine learning
- To predict the most suitable crop based on soil nutrients and environmental conditions

Dataset Description

A simulated dataset is used containing:

- **Nitrogen (N)** → Soil nutrient
- **Phosphorus (P)** → Root development
- **Potassium (K)** → Plant health
- **Temperature** → Climate condition
- **Humidity** → Moisture level
- **pH** → Soil acidity/alkalinity
- **Crop** → Target variable

Methodology

1. Data Preparation

The dataset is created using pandas and divided into:

- Features (input variables)
- Target (crop type)

2. Model Selection

A **Decision Tree Classifier** is used because:

- Easy to understand
- Works well with small datasets
- Suitable for classification problems

3. Model Training

The model is trained using the dataset to learn patterns between soil conditions and crop types.

4. Prediction

New input values are given to the model to predict the most suitable crop.

Results and Observations

- The model successfully predicts crops based on input conditions
- It learns patterns such as:
  - High humidity → Rice
  - Moderate temperature → Wheat
- The system provides quick and useful recommendations

Discussion

This system can help:

- Farmers choose the right crop
- Improve agricultural productivity
- Reduce losses due to wrong crop selection

Even though the dataset is small, it demonstrates how ML can be applied in agriculture.

Conclusion

The crop recommendation system shows how machine learning can assist in smart farming. By analyzing soil and environmental factors, the model can suggest suitable crops, making farming more efficient and data-driven.

Future Improvements

- Use real agricultural datasets
- Apply advanced models (Random Forest, SVM)
- Build a web/mobile application
- Include rainfall and seasonal data